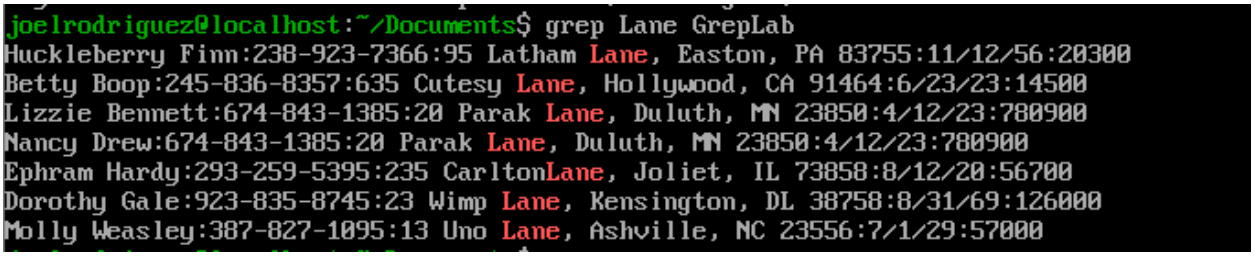
Joel Rodriguez
Fall 2026
CIS 245
Grep Lab

WHAT IS GREP?

Grep is Linux command that lets you search through a file for specific words or text. So, to avoid having to open a file and search the whole file and finding a certain text yourself, you can just tell grep what you're looking for and it will find it for you.

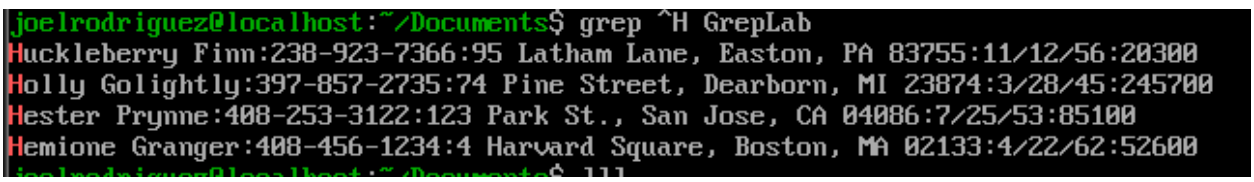
1. Print all lines containing the string Lane - **COMMAND**: grep Lane GrepLab




Source: your youtube video.

This command works because we used the command grep and used the word that we needed to find "Lane" and the name of the file that we're searching through "GrepLab." So we type Grep first, the word we want to search for and then finally the name of the file. The GrepLab file gets searched and it will display every line that contains the word Lane in it.

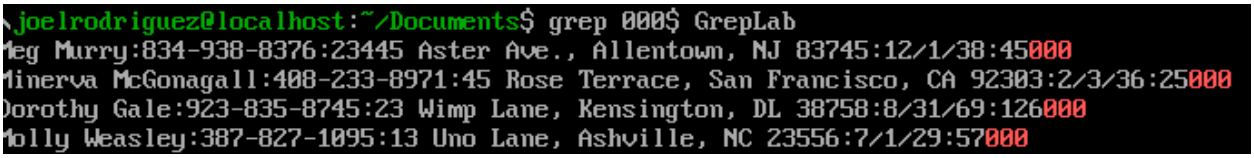
2. Print all lines where the person's first name starts with H - **COMMAND**: grep ^H GrepLab



Source: your video

I used this command to find people's names that start with the letter H. I noticed that all the names are in the beginning of the line. So, this command works because the ^ symbol means the beginning of the line. The H tells grep which letter it needs to look for. And ^H is telling grep to look for any line that starts with H in the GrepLab file.

3. Print all lines ending in three zeros (000) - **COMMAND**: grep 000$ GrepLab




I used this command because we're looking for lines that END with 000. The $ symbol is going to tell grep to look for three 0's at the end of the line. So, when I add 000$ we're telling grep to only display lines that end with 000 from the GrepLab file.

4. Print all lines that don't contain 408 - COMMAND: grep -v 408 GrepLab

```
Huckleberry Finn:238-923-7366:95 Latham Lane, Easton, PA 83755:11/12/56:20300

Betty Boop:245-836-8357:635 Cutesy Lane, Hollywood, CA 91464:6/23/23:14500

Dorian Gray:385-375-8395:3567 Populus Place, Caldwell, NJ 23875:6/18/68:23400

Holly Golightly:397-857-2735:74 Pine Street, Dearborn, MI 23874:3/28/45:245700

Ebenezer Scrooge:548-834-2348:583 Laurel Ave., Kingsville, TX 83745:10/1/35:58900

Westley Pirate:284-758-2857:23 Edgecliff Place, Lincoln , NB 92743:7/25/53:85100

Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200

Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200

Lizzie Bennett:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900

Nancy Drew:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900

Jo March:327-832-5728:3465 Mirlo Street, Peabody, MA 34756:10/2/65:35200

Victor Frankenstein:835-365-1284:454 Easy Street, Decatur, IL 75732:2/28/53:123500

Ephram Hardy:293-259-5395:235 CarltonLane, Joliet, IL 73858:8/12/20:56700

Meg Murry:834-938-8376:23445 Aster Ave., Allentown, NJ 83745:12/1/38:45000

Lucy Pevensie:385-573-8326:832 Ponce Drive, Gzary, IN 83756:12/1/46:268500

Mary Poppins:846-836-2837:6937 Ware Road, Milton, PA 93756:9/21/46:43500

Sir Lancelot:837-835-8257:474 Camelot Boulevard, Bath, WY 28356:5/13/69:24500

Zippy Pinhead:834-823-8319:2356 Bizarro Ave., Farmount, IL 84357:1/1/67:89500

Dorothy Gale:923-835-8745:23 Wimp Lane, Kensington, DL 38758:8/31/69:126000

Popeye Sailor:156-454-3322:945 Bluto Street, NotHere, USA 29358:3/19/35:22350

Luna Lovegood:385-898-8357:38 Fife Way, Abilene, TX 39673:1/5/58:95600

Molly Weasley:387-827-1095:13 Uno Lane, Ashville, NC 23556:7/1/29:57000

Arya Stark:438-910-7449:8235 Maple Street, Wilmington, VM 29085:9/23/63:68900
```

Source: Your video

Here we are using grep to look for 408 and hide it so we can display everything else. If we would've done grep 408 GrepLab the display would've shown every line that contains 408. Since we added the -v before the 408, no grep is looking for every line that contains 408 and make sure that It doesn't get shown in the display and it will show every line that doesn't have 408 in it.

5. Print all lines where birthdays are in the year 1935 (be careful of the date format! it's MM/DD/YY) - COMMAND: grep “/35\b” GrepLab

```
joelrodriguez@localhost:~/Documents$ grep /35\b GrepLab
joelrodriguez@localhost:~/Documents$ grep 35\b GrepLab
joelrodriguez@localhost:~/Documents$ grep "35\b" GrepLab
Betty Boop:245-836-8357:635 Cutesy Lane, Hollywood, CA 91464:6/23/23:14500
Holly Golightly:397-857-2735:74 Pine Street, Dearborn, MI 23874:3/28/45:245700
Ebenezer Scrooge:548-834-2348:583 Laurel Ave., Kingsville, TX 83745:10/1/35:58900
Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200
Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200
Victor Frankenstein:835-365-1284:454 Easy Street, Decatur, IL 75732:2/28/53:123500
Ephram Hardy:293-259-5395:235 CarltonLane, Joliet, IL 73858:8/12/20:56700
Sir Lancelot:837-835-8257:474 Camelot Boulevard, Bath, WY 28356:5/13/69:24500
Dorothy Gale:923-835-8745:23 Wimp Lane, Kensington, DL 38758:8/31/69:126000
Popeye Sailor:156-454-3322:945 Bluto Street, NotHere, USA 29358:3/19/35:22350
Arya Stark:438-910-7449:8235 Maple Street, Wilmington, VM 29085:9/23/63:68900
joelrodriguez@localhost:~/Documents$ grep "/35\b" GrepLab
Ebenezer Scrooge:548-834-2348:583 Laurel Ave., Kingsville, TX 83745:10/1/35:58900
Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200
Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200
Popeye Sailor:156-454-3322:945 Bluto Street, NotHere, USA 29358:3/19/35:22350
joelrodriguez@localhost:~/Documents$
```

Source: https://stackoverflow.com/questions/39443226/meaning-of-b-in-the-below-regex-using-grep AND PDF File I found online.

This is where things start getting tricky. We had to start using years and learn how to separate the years from the other numbers. So, I used this command to find all the lines where the birthday is in 1935. Since these dates are written in the MM/DD/YY format the /35 will look for birthdays that end in 35 and not any other number that contains the number 35 in it. The \b is a word boundary, so this makes sure that 35 is treated as the end of the part of the date. Make sure when you run this command that the word boundary in quotation marks and this tells grep exactly what to look for.

6. Print all lines where the phone number is in an area code that starts with an 8 - **COMMAND**: grep ":[8][0-9][0-9]-" GrepLab

```
joelrodriguez@localhost:~/Documents$ grep ":[8][0-9][0-9]-" GrepLab
Victor Frankenstein:835-365-1284:454 Easy Street, Decatur, IL 75732:2/28/53:123500
Meg Murry:834-938-8376:23445 Aster Ave., Allentown, NJ 83745:12/1/38:45000
Mary Poppins:846-836-2837:6937 Ware Road, Milton, PA 93756:9/21/46:43500
Sir Lancelot:837-835-8257:474 Camelot Boulevard, Bath, WY 28356:5/13/69:24500
Zippy Pinhead:834-823-8319:2356 Bizarro Ave., Farmount, IL 84357:1/1/67:89500
```

Sources: PDF File

Notice how all the area codes have a ":" right before it? We're going to use that to help us grab the lines that where the area codes start with 8. So, the colon will tell grep where to start. [8] this tells grep that if the number starts with 8 that it should keep it and go on the next number. Each [0-9] will tell grep that aslong that it's a number after 8 that it should fall in grep command. And the dash at the end makes sure that the 3 numbers that are being recorded ([8][0-9][0-9]) needs to be followed up with a dash.

7. Print all lines containing an uppercase letter, followed by 4 lowercase, a space and one uppercase letter. - **COMMAND**: grep -E "[A-Z][a-z]{4} [A-Z]" GrepLab

```
joelrodriguez@localhost:~/Documents$ grep -E "[A-Z][a-z]{4} [A-Z]" GrepLab
Betty Boop:245-836-8357:635 Cutesy Lane, Hollywood, CA 91464:6/23/23:14500
Holly Golightly:397-857-2735:74 Pine Street, Dearborn, MI 23874:3/28/45:245700
Lizzie Bennett:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900
Nancy Drew:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900
Jo March:327-832-5728:3465 Mirlo Street, Peabody, MA 34756:10/2/65:35200
Meg Murry:834-938-8376:23445 Aster Ave., Allentown, NJ 83745:12/1/38:45000
Lucy Pevensie:385-573-8326:832 Ponce Drive, Gzary, IN 83756:12/1/46:268500
Zippy Pinhead:834-823-8319:2356 Bizarro Ave., Farmount, IL 84357:1/1/67:89500
Popeye Sailor:156-454-3322:945 Bluto Street, NotHere, USA 29358:3/19/35:22350
Daenerys Targaryen:408-724-0140:1222 Oxbow Court, Sunnyvale, CA 94087:5/19/66:34200
Molly Weasley:387-827-1095:13 Uno Lane, Ashville, NC 23556:7/1/29:57000
Arya Stark:438-910-7449:8235 Maple Street, Wilmington, VM 29085:9/23/63:68900
```

Source: PDF File

Here is where we're going to start using an extended regular expression. An Extended regular expression is a way of searching for patterns instead of searching for an exact word. And that's why I used -E here. -E represents

extended regular expression. Grep is first going to look for a capital letter cause I added [A-Z] and then it's going to look for four straight lower case letters because I added [a-z]{4}. {4} is telling grep to look for 4 straight patterns of what was inputted right before it. Make sure to add a space and then [A-Z].

8. Print lines where the address begins with a two or three digit number - **COMMAND**: grep -E "\b[0-9]{2,3} " GrepLab

```
joelrodriguez@localhost:~/Documents$ grep -P "^[^:]+:[^:]+:\d{2,3}\s" GrepLab
Huckleberry Finn:238-923-7366:95 Latham Lane, Easton, PA 83755:11/12/56:20300
Betty Boop:245-836-8357:635 Cutesy Lane, Hollywood, CA 91464:6/23/23:14500
Holly Golightly:397-857-2735:74 Pine Street, Dearborn, MI 23874:3/28/45:245700
Ebenezer Scrooge:548-834-2348:583 Laurel Ave., Kingsville, TX 83745:10/1/35:58900
Hester Prynne:408-253-3122:123 Park St., San Jose, CA 04086:7/25/53:85100
Westley Pirate:284-758-2857:23 Edgecliff Place, Lincoln , NB 92743:7/25/53:85100
Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200
Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200
Lizzie Bennett:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900
Nancy Drew:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900
Victor Frankenstein:835-365-1284:454 Easy Street, Decatur, IL 75732:2/28/53:123500
Ephram Hardy:293-259-5395:235 CarltonLane, Joliet, IL 73858:8/12/20:56700
Lucy Pevensie:385-573-8326:832 Ponce Drive, Gzary, IN 83756:12/1/46:268500
Sir Lancelot:837-835-8257:474 Camelot Boulevard, Bath, WY 28356:5/13/69:24500
Minerva McGonagall:408-233-8971:45 Rose Terrace, San Francisco, CA 92303:2/3/36:25000
Dorothy Gale:923-835-8745:23 Wimp Lane, Kensington, DL 38758:8/31/69:126000
Popeye Sailor:156-454-3322:945 Bluto Street, NotHere, USA 29358:3/19/35:22350
Luna Lovegood:385-898-8357:38 Fife Way, Abilene, TX 39673:1/5/58:95600
Molly Weasley:387-827-1095:13 Uno Lane, Ashville, NC 23556:7/1/29:57000
```

Source: PDF FIle

This command is looking for addresses that begin with two or three digit number. This command has a word boundary (/b), [0-9] tells grep that aslong as the it's a number that it's a good start and {2,3} tells grep that there must be atleast 2 digits but no more than three. The space makes sure that the number of the address gets followed up by a space since all the numbers in the addresses are followed by that and it will make sure that it doesn't add any unnecessary lines.

9. Print lines where the person lives in Mass or Illinois - COMMAND: grep -E "MA|IL" GrepLab

```
joelrodriguez@localhost:~/Documents$ grep -E "MA|IL" GrepLab
Jo March:327-832-5728:3465 Mirlo Street, Peabody, MA 34756:10/2/65:35200
Victor Frankenstein:835-365-1284:454 Easy Street, Decatur, IL 75732:2/28/53:123500
Ephram Hardy:293-259-5395:235 CarltonLane, Joliet, IL 73858:8/12/20:56700
Hemione Granger:408-456-1234:4 Harvard Square, Boston, MA 02133:4/22/62:52600
Zippy Pinhead:834-823-8319:2356 Bizarro Ave., Farmount, IL 84357:1/1/67:89500
joelrodriguez@localhost:~/Documents$
```

Source: your video

This command finds all the lines of which people live in Massachusetts and Illinois. I know the | will divide the two states and it's telling grep to find lines that live in both states. Spelled it out fully before I realized that I had to use the initials because that's how its written in the file.

10. Print lines containing the addresses that aren't on a street (does not contain St or Street, case-insensitive) - **COMMAND**: grep -ivE "St|Street" GrepLab

```
joelrodriguez@localhost:~/Documents$ grep -ivE "St|Street" GrepLab
Betty Boop:245-836-8357:635 Cutesy Lane, Hollywood, CA 91464:6/23/23:14500
Dorian Gray:385-375-8395:3567 Populus Place, Caldwell, NJ 23875:6/18/68:23400

Ebenezer Scrooge:548-834-2348:583 Laurel Ave., Kingsville, TX 83745:10/1/35:58900



Lizzie Bennett:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900
Nancy Drew:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900

Ephram Hardy:293-259-5395:235 CarltonLane, Joliet, IL 73858:8/12/20:56700

Lucy Pevensie:385-573-8326:832 Ponce Drive, Gzary, IN 83756:12/1/46:268500

Mary Poppins:846-836-2837:6937 Ware Road, Milton, PA 93756:9/21/46:43500
Sir Lancelot:837-835-8257:474 Camelot Boulevard, Bath, WY 28356:5/13/69:24500
Minerva McGonagall:408-233-8971:45 Rose Terrace, San Francisco, CA 92303:2/3/36:25000
Zippy Pinhead:834-823-8319:2356 Bizarro Ave., Farmount, IL 84357:1/1/67:89500
Dorothy Gale:923-835-8745:23 Wimp Lane, Kensington, DL 38758:8/31/69:126000

Luna Lovegood:385-898-8357:38 Fife Way, Abilene, TX 39673:1/5/58:95600
Daenerys Targaryen:408-724-0140:1222 Oxbow Court, Sunnyvale, CA 94087:5/19/66:34200
Molly Weasley:387-827-1095:13 Uno Lane, Ashville, NC 23556:7/1/29:57000
```

Source: your video

This command finds lines containing addresses that are not on a street. The - i makes the search case-sensitive, so whether it's capital or lowercase it should not matter. The -v makes sure to search and displays the lines that DO NOT contain this pattern. And the -E enables the extended regular expression so we can be able to use |. So anything that has St or Street should not be included into the display.